# Research Directions for Variable Selection in Wasserstein Robust Testing

## 1 Starting point

The theory note writes the selection problem $\min_{|S|\le d} V(S)$ as the minimization of a convex function $F$ over $\mathcal{Z}_d$ and solves it by outer approximation. In development checks with the hinge loss, the method evaluated every subset before it closed the gap, whatever cuts it used, and the continuous relaxation was weak: on an instance with $D = 10$ and $d = 2$, $F$ takes the value 0.58 at the uniform point $(d/D)\mathbf{1}$, against the optimum 1.40.

This note starts from one question: does the weakness belong to the formulation or to the problem? The answer developed below is that it belongs to the problem and has a plain meaning. *Mixing coordinates helps the defender*: a detector that may weigh all coordinates, or randomize over them, separates the classes better than any fixed sparse projection. Whether this help is real depends on whom the defender faces, and the three answers organize the note.

- Against sampling error the help is spurious. In high dimension a robust test certifies separation between identical classes; selection with a radius that grows with the selected dimension removes the effect (Direction A, Section 5).
- In the optimization the help is the relaxation gap, and it is why the problem is hard (Section 4; Direction B, Section 6).
- Against an adversary who must split one transport budget across coordinates, the help is genuine, and randomized sparse tests collect it (Direction C, Section 7).

Section 2 records how the argument was found, including the routes that failed. Section 3 gives the reformulation on which everything rests. Each direction then lists its reference theory, a derivation route and its target results, and every statement is marked as proved, sketched or conjectured. Section 8 sets priorities.

Notation follows the theory note. For a distribution $Q$ on $\mathbb{R}^D$, $Q[S]$ is its image under $x \mapsto x[S]$, and every transport cost uses the $\ell_1$ norm on the coordinates in question. The hinge loss is $\ell(t) = (1 + t)_+$.

## 2 Line of thought

The argument came in five steps, each answering the question that the previous one raised.

**1. What does $V(S)$ measure?** A detector in $\mathcal{F}_S$ reads only $x[S]$, and projection maps each Wasserstein ball onto the ball of the same radius around the projected sample. So $V(S)$ depends only on the projected samples and the radii. At fixed distributions the smallest risk over detectors is a classical object: each loss defines an $f$-divergence [20, 27]. Hence $V(S)$ is 2 minus the smallest divergence the adversary can reach (Section 3.1). This is Huber's picture of robust testing [15, 16], with Wasserstein balls in place of contamination neighbourhoods [12].

**2. What does the adversary do?** For the hinge loss the divergence is total variation, and reducing it means creating overlap. Overlap is paid for by moving mass of one class onto mass of the other, so the adversary solves a partial transport problem, a budgeted relative of Strassen's coupling theorem [30]. On samples it is a bipartite matching (Section 3.2).

**3. Where does the selection enter?** The matching dual prices transport at a rate $\lambda$ and leaves a bounded $\lambda$-Lipschitz witness, so the robust test is a bounded-Lipschitz two-sample statistic (Section 3.3). The selection enters only through the pair costs, which are linear in $z = \mathbf{1}_S$. The whole problem is therefore one compact mixed-integer program, and its relaxation replaces a set of coordinates by a weighted $\ell_1$ metric (Section 4).

**4. Why is the relaxation weak?** A weighted metric averages coordinates. Averaging concentrates the cross-class pair costs around their mean, and the adversary loses the few cheap pairs that it can always find in a sparse projection. This is the mechanism "mixing helps the defender". The relaxation gap measures it, and NP-hardness shows that no formulation removes it in the worst case.

**5. Is the help real?** It depends on the adversary, which gives the three directions of Section 1.

**Routes that failed.** (a) Perspective strengthening, which makes sparse regression easy [2], relies on a strongly convex ridge term. Here $z$ enters only through the linear cost $\sum_m z_m|x[m] - y[m]|$, whose perspective is itself, and the relaxation is already the weighted-metric problem. (b) Bounding each pair cost by its $d$ largest coordinate differences gives a valid bound, but each pair then uses its own coordinates, and the bound is never stronger than the relaxation (Section 4). (c) The randomized value of Direction C is a lower bound for the fixed problem, but on every instance of Table 3 it lies below the relaxation. (d) Cut engineering for outer approximation, such as least favorable cuts, Pareto cuts or several tangents per evaluation, reduced oracle time by 30–45% in development checks, but cannot close a gap that belongs to the problem. (e) Sinkhorn ambiguity sets place the selection inside $\exp(-c/\varepsilon)$ and lose the matching structure; neural generators of least favorable distributions lose the certificate.

# 3 The reformulation

## 3.1 The robust value is a divergence

For $a, b \geq 0$ let $\psi(a, b) = \inf_{t\in\mathbb{R}}\{a\ell(-t) + b\ell(t)\}$. It is concave and positively homogeneous, and $\psi(a, a) = 2a$ because convexity gives $\ell(-t) + \ell(t) \geq 2\ell(0) = 2$. Hence

$$D_\ell(Q_0, Q_1) = 2 - \int \psi(dQ_0, dQ_1)$$

is an $f$-divergence between probability distributions. For $\ell(t) = (1 + t)_+$, $e^t$ and $\log_2(1 + e^t)$ it equals 2 TV, the squared Hellinger distance $\int(\sqrt{dQ_0} - \sqrt{dQ_1})^2$, and twice the Jensen–Shannon divergence in bits. Write $\mathcal{P}_k[S] = \{Q : \mathcal{W}_1(\widehat{\mathbb{P}}_k[S], Q) \leq \rho_k\}$.

**Proposition 3.1** (Divergence form; proved)**.** *For every $S \subseteq [D]$, $V(S) = 2 - \min_{Q_k\in\mathcal{P}_k[S]} D_\ell(Q_0, Q_1)$.*

*Proof.* Fix $Q_0$ and $Q_1$. Minimizing the risk $\mathbb{E}_{Q_0}\ell(-f) + \mathbb{E}_{Q_1}\ell(f)$ point by point over $f \in \mathcal{F}_S$ gives $\int \psi(dQ_0[S], dQ_1[S])$. Projection maps the ambiguity sets into $\mathcal{P}_k[S]$, and every element of $\mathcal{P}_k[S]$ lifts with the same cost. Exchanging the minimum over $f$ and the supremum over distributions therefore gives $V(S) \geq 2 - \inf D_\ell$, and Lemma 2.1 of the theory note attains this bound with distributions supported on the projected samples. □

**Reference theory.** The correspondence between losses and divergences is due to Nguyen, Wainwright and Jordan [20] and is developed by Reid and Williamson [27]. Least favorable pairs for robust tests go back to Huber [15] and Huber and Strassen [16]; Wasserstein balls are those of [12]. Selecting a projection that maximizes a distance between two samples is the theme of principal differences analysis [19], projected Wasserstein tests [33], subspace robust distances [23], feature-robust transport [24] and kernel variable selection [32]. Here the distance is a divergence measured after a transport attack, and its value is the risk certificate of a minimax test. The mirror construction, a Wasserstein distance measured after a total-variation attack, is the robust transport of [21].

## 3.2 Hinge loss: a budgeted matching

For the hinge loss the divergence is total variation, and the attack becomes combinatorial. Write $C^m_{ij} = |x^{(0)}_i[m] - x^{(1)}_j[m]|$ and $c_{ij}(S) = \sum_{m\in S} C^m_{ij}$, and let

$$m_S(\rho) = \max_{\pi\geq 0}\Big\{\sum_{i,j}\pi_{ij} : \sum_j \pi_{ij} \leq \frac{1}{n_0},\ \sum_i \pi_{ij} \leq \frac{1}{n_1},\ \sum_{i,j}\pi_{ij}c_{ij}(S) \leq \rho\Big\} \tag{1}$$

be the largest cross-class mass that can be paired within a total transport budget $\rho$.

**Proposition 3.2** (Matching form; proved)**.** *For the hinge loss,* $V(S) = 2\, m_S(\rho_0 + \rho_1)$.

*Proof.* By Proposition 3.1, $V(S)/2 = 1 - \min \mathrm{TV}(Q_0, Q_1)$ is the largest overlap $\int \min(dQ_0, dQ_1)$ of two distributions in the projected balls. Let transport plans $\gamma_0, \gamma_1$ reach $Q_0, Q_1$ from the projected samples. At each point, split the overlap by the source of its class-0 mass and, independently, by the source of its class-1 mass. This gives masses $\pi_{ij}$ with row sums at most $1/n_0$, column sums at most $1/n_1$, and total equal to the overlap. A unit of $\pi_{ij}$ reaches a common point from $x_i^{(0)}[S]$ and from $x_j^{(1)}[S]$, so by the triangle inequality it costs the two plans together at least $c_{ij}(S)$. Hence the overlap is at most $m_S(\rho_0+\rho_1)$. Conversely, let $\pi$ be feasible with $\rho = \rho_0 + \rho_1$, set $\theta = \rho_0/\rho$, and send mass $\pi_{ij}$ of each class to the point $x_i^{(0)}[S] + \theta(x_j^{(1)}[S] - x_i^{(0)}[S])$. Class 0 pays $\theta c_{ij}(S)$ per unit and class 1 pays $(1-\theta)c_{ij}(S)$, so both budgets hold, and the overlap is at least $\sum_{i,j} \pi_{ij}$. □

Three consequences are used below. First, the radii enter only through $\rho_0 + \rho_1$; on random instances with unequal radii the solver of the theory note agrees with $2m_S$ to $10^{-10}$. Second, the least favorable distributions pair up cross-class points: the adversary is a matchmaker with a transport budget, and $S$ separates the classes when few cross-class pairs are cheap in the coordinates of $S$. Third, evaluating $V(S)$ is a bipartite matching with one side constraint and $n_0 n_1$ variables, instead of a transport plan over $N^2$ pairs with two budgets.

The proof never uses that the distributions are empirical. For any $P_0, P_1$, the largest overlap reachable within budgets $r_0, r_1$ is the largest mass of a partial coupling of $P_0[S]$ and $P_1[S]$ whose transport cost is at most $r_0 + r_1$. This is the budget form of optimal partial transport [8, 3, 4], and it plays for Wasserstein balls the role that Strassen's theorem [30] plays for Prokhorov neighbourhoods: it turns a statement about perturbed distributions into one about couplings.

### 3.3 The dual: bounded Lipschitz witnesses

For $\lambda \geq 0$ and distributions $Q_0, Q_1$ on $\mathbb{R}^{|S|}$, let

$$\beta_\lambda(Q_0, Q_1) = \sup\big\{\mathbb{E}_{Q_0} h - \mathbb{E}_{Q_1} h :\ |h| \leq \tfrac{1}{2},\ |h(\xi) - h(\xi')| \leq \lambda \|\xi - \xi'\|_1\big\}.$$

This is a variant of Dudley's bounded-Lipschitz distance [7] with the Lipschitz constant set to $\lambda$. It increases from 0 at $\lambda = 0$ to $\mathrm{TV}(Q_0, Q_1)$ as $\lambda \to \infty$, and it is concave in $\lambda$, since $th_1 + (1-t)h_2$ is admissible for $t\lambda_1 + (1-t)\lambda_2$.

**Proposition 3.3** (Lipschitz witness; proved)**.** *For the hinge loss,*

$$V(S) = 2 - 2 \max_{\lambda \geq 0} \big\{\beta_\lambda\big(\widehat{\mathbb{P}}_0[S], \widehat{\mathbb{P}}_1[S]\big) - \lambda(\rho_0 + \rho_1)\big\},$$

*and if* $(\lambda, h)$ *attains the maximum, the detector* $f(x) = 2h(x[S])$ *is optimal:* $R(f) = V(S)$.

*Proof.* Write $\rho = \rho_0 + \rho_1$. Dualizing only the budget in Eq. (1) gives $m_S(\rho) = \min_{\lambda \geq 0}\{\lambda\rho + M(\lambda)\}$, where $M(\lambda) = \max_\pi \sum_{i,j} \pi_{ij}(1 - \lambda c_{ij}(S))$ over the row and column constraints. The dual of $M(\lambda)$ is $\min\{\bar{u} + \bar{v} :\ u_i + v_j \geq 1 - \lambda c_{ij}(S),\ u, v \geq 0\}$, with $\bar{u} = \frac{1}{n_0}\sum_i u_i$ and $\bar{v} = \frac{1}{n_1}\sum_j v_j$. Put $h_i = \frac{1}{2} - u_i$ at the class-0 points and $h'_j = v_j - \frac{1}{2}$ at the class-1 points. Then $1 - M(\lambda)$ is the largest value of $\frac{1}{n_0}\sum_i h_i - \frac{1}{n_1}\sum_j h'_j$ subject to $h_i - h'_j \leq \lambda c_{ij}(S)$, $h_i \leq \frac{1}{2}$ and $h'_j \geq -\frac{1}{2}$. Raising any $h_i < -\frac{1}{2}$ to $-\frac{1}{2}$, or lowering any $h'_j > \frac{1}{2}$ to $\frac{1}{2}$, keeps feasibility and does not decrease the objective. The clipped function $h(\xi) = \max\{-\frac{1}{2}, \min\{\frac{1}{2}, \min_j(h'_j + \lambda\|x_j^{(1)}[S] - \xi\|_1)\}\}$ is then admissible in $\beta_\lambda$, is at least $h_i$ at the class-0 points and at most $h'_j$ at the class-1 points. Conversely every admissible $h$ gives feasible values, so $1 - M(\lambda) = \beta_\lambda$, and Proposition 3.2 gives the formula. For the detector, $|2h| \leq 1$ makes the hinge losses linear, and Kantorovich–Rubinstein duality gives, for every admissible pair $(Q_0, Q_1)$,

$$\mathbb{E}_{Q_0}(1 - 2h) + \mathbb{E}_{Q_1}(1 + 2h) \leq 2 - 2\big(\mathbb{E}_{\widehat{\mathbb{P}}_0} h - \mathbb{E}_{\widehat{\mathbb{P}}_1} h\big) + 2\lambda\rho = 2 - 2\big(\beta_\lambda - \lambda\rho\big) = V(S). \qquad \square$$

On random instances the formula agrees with Eq. (1) to $10^{-8}$. The proposition gives the robust hinge test a plain reading: it is a bounded-Lipschitz two-sample statistic whose Lipschitz constant is priced at the total radius. A large $\lambda$ lets the witness separate the samples but pays $\lambda\rho$; the radius decides how

smooth an honest witness must be. Variable selection chooses the $d$ coordinates on which this penalized distance is largest, the analogue of selecting coordinates for a kernel two-sample test [32]. The function $1 - M(\lambda)$ is also an unbalanced transport cost in which unmatched mass is paid at a fixed price, the generalized Wasserstein distance of [25], whose dual is the bounded-Lipschitz norm [14]. Finally, $m_S$ is the minimum of the tangents $\rho \mapsto \lambda\rho + M(\lambda)$, so it is concave and nondecreasing in $\rho$ with slope equal to the optimal price; Section 7 uses these tangents.

## 4 Relaxation and hardness

Linear programming duality for Eq. (1) gives

$$m_S(\rho) = \min_{u,v,\lambda \geq 0} \left\{ \rho\lambda + \bar{u} + \bar{v} : \; u_i + v_j + \lambda c_{ij}(S) \geq 1 \right\}.$$

Each cross-class pair is either paid for by its endpoints or separated by its priced distance $\lambda c_{ij}(S)$, and selecting $S$ decides which coordinates the adversary pays for. With $w_m = \lambda z_m$ and $\rho = \rho_0 + \rho_1$, the whole selection problem becomes one compact program:

$$\begin{aligned} \min_{z \in \mathcal{Z}_d,\, u,v,w,\lambda \geq 0} \quad & \rho\lambda + \bar{u} + \bar{v} \\ \text{subject to} \quad & u_i + v_j + \sum_m w_m C^m_{ij} \geq 1, \qquad w_m \leq \lambda, \qquad \rho w_m \leq z_m, \qquad \sum_m w_m \leq d\lambda. \end{aligned} \tag{2}$$

**Proposition 4.1** (Compact program; proved)**.** *The optimal value of Eq. (2) is* $\min_{|S| \leq d} m_S(\rho) = \frac{1}{2} \min_{|S| \leq d} V(S)$. *Its continuous relaxation, with* $\mathcal{Z}_d$ *replaced by* $\{z \in [0,1]^D : \mathbf{1}^\top z \leq d\}$, *has the value* $\min_\zeta m_\zeta(\rho)$ *over the same set, where* $m_\zeta$ *is* $m_S$ *with the cost* $c_{ij}(S)$ *replaced by the weighted cost* $\sum_m \zeta_m C^m_{ij}$.

*Proof.* The point $u = \mathbf{1}$, $v = 0$, $\lambda = 0$ has value 1, so $\rho\lambda \leq 1$ at an optimum. For $z = \mathbf{1}_S$ the constraints force $w_m = 0$ outside $S$ and allow $w_m = \lambda$ inside, which is optimal because $C^m_{ij} \geq 0$; the value is the dual of $m_S(\rho)$. For the relaxation, a feasible point with $\lambda > 0$ gives $\zeta = w/\lambda$ in the relaxed set and the dual objective of $m_\zeta(\rho)$; with $\lambda = 0$ the rows give $\bar{u} + \bar{v} \geq 1$. Conversely, for $\zeta$ in the relaxed set and an optimal dual $(u, v, \lambda)$ of $m_\zeta(\rho)$, the point $z = \zeta$, $w = \lambda\zeta$ is feasible because $\rho\lambda \leq 1$. □

The relaxation lets the defender choose a weighted $\ell_1$ metric instead of a set of coordinates, and weighted metrics are genuinely better defenses. Averaging coordinates concentrates the pair costs $\sum_m \zeta_m C^m_{ij}$ around their mean, and the adversary loses the cheap pairs it finds in a sparse projection. Table 1 shows the effect: the relaxed value is one half to two thirds of the optimum. It is still stronger than the relaxation of the theory note, whose $F$ already takes the value 0.58 at the uniform point on the first instance. Since every relaxed cost satisfies $\sum_m \zeta_m C^m_{ij} \leq T_{ij}$, the sum of the $d$ largest $C^m_{ij}$, and $m$ decreases with the costs, the pairwise bound $m_T(\rho)$ of Section 2(b) never exceeds the relaxation.

| $D$ | $N$ | $d$ | $\min_S V(S)$ | relaxation | Eq. (2) by HiGHS | enumeration |
|---|---|---|---|---|---|---|
| 10 | 30 | 2 | 1.40 | 0.89 | 0.2 s, 17 nodes | 0.2 s |
| 15 | 30 | 2 | 1.12 | 0.75 | 0.4 s, 25 nodes | 0.5 s |
| 12 | 30 | 3 | 0.87 | 0.49 | 0.7 s, 302 nodes | 1.1 s |
| 30 | 40 | 3 | 0.75 | 0.43 | 26 s, 4.1k nodes | 22 s |
| 50 | 40 | 3 | 0.80 | 0.40 | 235 s, 26k nodes | – |
| 100 | 40 | 3 | $\leq 0.88$ | 0.37 | time limit 600 s | – |

Table 1: Hinge loss, $\rho_0 + \rho_1 = 0.6$, values on the scale of $V$ (twice the value of Eq. (2)). Both classes are standard normal except that class 1 is shifted by 0.8 in coordinates 1–3 and scaled by 1.8 in coordinate 4. Program Eq. (2) reproduces the enumerated optimum wherever enumeration was run. Development check, not a formal result.

The combinatorial core is also hard in the worst case.

**Proposition 4.2** (Hardness; proved)**.** *Let the data be binary and* $\rho_0 + \rho_1 < \min\{1/n_0, 1/n_1\}$. *Then* $\min_{|S|\le d} V(S) \le 2(\rho_0 + \rho_1)$ *if and only if some* $S$ *with* $|S| \le d$ *separates every cross-class pair, that is,* $x_i^{(0)}[S] \ne x_j^{(1)}[S]$ *for all* $i, j$. *Deciding this is the* $k$*-feature-set problem, which is NP-complete [6] and W[2]-complete in* $d$ *[5]. Hence computing* $\min_{|S|\le d} V(S)$ *is NP-hard.*

*Proof.* Write $\rho = \rho_0 + \rho_1$. If $S$ separates every pair, then $c_{ij}(S) \ge 1$ for all pairs and $m_S(\rho) \le \rho$. Otherwise some pair has $c_{ij}(S) = 0$ and can be matched at no cost, so $m_S(\rho) \ge \min\{1/n_0, 1/n_1\} > \rho$. Proposition 3.2 completes the proof. □

Diminishing returns do not rescue greedy selection either. With class 0 at $(0,0), (1,1)$ and class 1 at $(0,1), (1,0)$, each coordinate alone leaves the projected classes identical, so $V = 2$, while the two coordinates together give $V = 2(\rho_0 + \rho_1)$ when $\rho_0 + \rho_1 \le 1$. A strong convex relaxation is therefore not to be expected, and progress must use information that the weighted-metric problem does not see: the sample size (Direction A), the matching structure (Direction B), or the value of mixing itself (Direction C).

# 5 Direction A: selection makes robust tests honest

## 5.1 The phenomenon

Table 2 draws both classes from the same distribution, so the correct robust value is 2. Yet $V([D])$ falls to 0.03 at $D = 50$. In high dimension the empirical points are isolated, every cross-class pair is expensive to match, and the robust test certifies a separation that does not exist. Selection limits the effect, but the best of many subsets is optimistic too, and in the first instance of Table 1 the optimal pair consists of two uninformative coordinates. A radius that ignores the dimension is not honest.

| $D$ | 1 | 2 | 3 | 5 | 10 | 20 | 50 |
|---|---|---|---|---|---|---|---|
| $V([D])$ | 1.98 | 1.66 | 1.17 | 0.59 | 0.21 | 0.09 | 0.03 |
| $\min_{\lvert S\rvert=2} V(S)$ | – | 1.66 | 1.65 | 1.53 | 1.39 | 1.40 | – |

Table 2: Both classes $N(0, I_D)$, $n_0 = n_1 = 15$, hinge loss, $\rho_0 + \rho_1 = 0.6$; mean over five draws. Development check.

**Reference theory.** Rates of empirical measures in $\mathcal{W}_1$ [10, 34] and the bounded-difference inequality [17] give coverage; the estimation of integral probability metrics [29] is the same question for the witness class of Proposition 3.3. Finite-sample Wasserstein robust optimization [11] shows when smaller radii suffice. Transport between distributions that differ only in a few directions is the spiked transport model [22], close to subspace methods [23, 33]. Choosing a model size by a data-dependent bound is structural risk minimization [1].

**Route.** Two deterministic facts suffice. The triangle inequality nests the empirical and population balls, which sandwiches $V(S)$ between population values (Lemma 5.1); and the detector certificate degrades linearly in the coverage error (Proposition 5.2). Probability enters once, through the event that every projected empirical distribution of dimension at most $d$ lies within $\varepsilon$ of its population. Its radius follows from a dyadic $\mathcal{W}_1$ bound in dimension $d$, a bounded-difference inequality and a union bound over the $\binom{D}{d}$ subsets (Lemma 5.4). Nearest-neighbour distances give a matching lower bound (Proposition 5.3).

**Population objects.** Let $\mathbb{P}_0, \mathbb{P}_1$ be the true distributions, $R_{\text{true}}(f) = \mathbb{E}_{\mathbb{P}_0}\ell(-f) + \mathbb{E}_{\mathbb{P}_1}\ell(f)$, and, for radii $r = (r_0, r_1)$,

$$V_{\text{pop}}(S; r) = 2 - \inf\big\{D_\ell(Q_0, Q_1) : \ \mathcal{W}_1(\mathbb{P}_k[S], Q_k) \le r_k, \ k = 0, 1\big\}, \qquad B_S = V_{\text{pop}}(S; 0).$$

By the proof of Proposition 3.1, $B_S = \min_{f \in \mathcal{F}_S} R_{\text{true}}(f)$ is the Bayes risk of the loss from the coordinates $S$; for the hinge loss $B_S = 2(1 - \text{TV}(\mathbb{P}_0[S], \mathbb{P}_1[S]))$ is twice the smallest total error of a test that reads $S$. $V_{\text{pop}}$ increases with $r$, and $s \mapsto V_{\text{pop}}(S; sr)$ is concave, because $D_\ell$ is jointly convex and the balls grow linearly in $s$. We write $V(S; \rho)$ for the empirical value with radii $\rho = (\rho_0, \rho_1)$.

## 5.2 Results

**Lemma 5.1** (Sandwich; proved)**.** *If* $\mathcal{W}_1(\widehat{\mathbb{P}}_k[S], \mathbb{P}_k[S]) \le \varepsilon_k \le \rho_k$ *for* $k = 0, 1$*, then*

$$B_S \le V_{\mathrm{pop}}(S; \rho - \varepsilon) \le V(S; \rho) \le V_{\mathrm{pop}}(S; \rho + \varepsilon).$$

*Proof.* By the triangle inequality, the ball of radius $\rho_k - \varepsilon_k$ around $\mathbb{P}_k[S]$ lies in the ball of radius $\rho_k$ around $\widehat{\mathbb{P}}_k[S]$, which lies in the ball of radius $\rho_k + \varepsilon_k$ around $\mathbb{P}_k[S]$. By Proposition 3.1 the robust value increases with the balls. □

**Proposition 5.2** (Certificate of the detector; proved)**.** *Let* $(g, a, \lambda)$ *solve the detector program of the theory note at S, and let* $f$ *be its extension in Appendix A there, used as the test that declares class* 1 *when* $f < 0$*. For any* $\mathbb{P}_0, \mathbb{P}_1$ *with finite first moments,*

$$\mathbb{P}_0(f < 0) + \mathbb{P}_1(f \ge 0) \le R_{\mathrm{true}}(f) \le V(S) + \sum_{k=0}^{1} \lambda_k \big(\mathcal{W}_1(\widehat{\mathbb{P}}_k[S], \mathbb{P}_k[S]) - \rho_k\big).$$

*Proof.* The first inequality holds because $\ell \ge 1$ on $[0, \infty)$. The envelope $E_k$ of Appendix A dominates the class-$k$ loss of $f$, is $\lambda_k$-Lipschitz on $\mathbb{R}^{|S|}$, and satisfies $E_k(x_i^{(k)}[S]) \le a_i^{(k)}$. Kantorovich–Rubinstein duality gives $\mathbb{E}_{\mathbb{P}_k} E_k \le \frac{1}{n_k} \sum_i a_i^{(k)} + \lambda_k \mathcal{W}_1(\widehat{\mathbb{P}}_k[S], \mathbb{P}_k[S])$. Summing over $k$ gives the second inequality. □

The next result shows that the radius must grow with the selected dimension.

**Proposition 5.3** (Radius lower bound; proved)**.** *Use the hinge loss, let* $\mathbb{P}_0 = \mathbb{P}_1$*, and let the projection on* $S$*,* $|S| = d$*, have a density bounded by* $M$*. With probability at least* $1 - e^{-n_0/8}$*, for every* $c \in (0, 1)$*,*

$$V(S) \ge 2 - c \quad \Longrightarrow \quad \rho_0 + \rho_1 \ge (1 - c)\, \frac{d}{4e}\, (4 M n_1)^{-1/d}.$$

*Proof.* Let $\delta_i$ be the $\ell_1$ distance from $x_i^{(0)}[S]$ to the nearest projected class-1 point. Given class 1, the $\delta_i$ are independent, and $\Pr(\delta_i < r) \le n_1 M (2r)^d / d!$, the volume of $n_1$ $\ell_1$ balls times $M$. Let $r^\star$ make this bound equal to $1/4$; $d! \ge (d/e)^d$ gives $r^\star \ge \frac{d}{2e}(4Mn_1)^{-1/d}$. By Hoeffding's inequality, with probability at least $1 - e^{-n_0/8}$ at least half of the $\delta_i$ are at least $r^\star$. If $V(S) \ge 2 - c$, then $m_S \ge 1 - c/2$ by Proposition 3.2, so at most $c/2$ of the class-0 mass is unmatched. The far points carry mass at least $1/2$, hence contribute matched mass at least $(1 - c)/2$, each unit costing at least $r^\star$. So $\rho_0 + \rho_1 \ge (1 - c) r^\star / 2$. □

For Table 2 with $D = 10$, where $M = (2\pi)^{-5}$ and $n_1 = 15$, the bound forbids $V([D]) \ge 1.5$ at $\rho_0 + \rho_1 = 0.6$ with probability at least 0.84, consistent with the observed mean 0.21. The order $d\, n^{-1/d}$ is also sufficient.

**Lemma 5.4** (Uniform coverage; standard, sketched)**.** *Let* $\mathbb{P}_0, \mathbb{P}_1$ *be supported in* $[0, 1]^D$*,* $3 \le d \le D$ *and* $\delta \in (0, 1)$*. With probability at least* $1 - \delta$*, for* $k = 0, 1$ *and all* $|S| \le d$*,*

$$\mathcal{W}_1(\widehat{\mathbb{P}}_k[S], \mathbb{P}_k[S]) \le \varepsilon_k := c\, d\, n_k^{-1/d} + d \sqrt{\frac{d \log(eD/d) + \log(2/\delta)}{2 n_k}},$$

*with an absolute constant* $c$*; the argument below gives* $c \le 9$*.*

*Proof sketch.* Projecting onto a subset contracts $\mathcal{W}_1$, so it suffices to take $|S| = d$. (i) Expectation. Partition $[0, 1]^d$ into the $2^{jd}$ dyadic cubes of side $2^{-j}$, whose $\ell_1$ diameter is $d 2^{-j}$. The dyadic bound of [34, Proposition 1] gives $\mathcal{W}_1 \le d\big(2^{-J} + \sum_{j=1}^{J} 2^{1-j} \sum_Q |\widehat{\mathbb{P}}(Q) - \mathbb{P}(Q)|\big)$. Since $\mathbb{E}|\widehat{\mathbb{P}}(Q) - \mathbb{P}(Q)| \le \sqrt{\mathbb{P}(Q)/n}$ and Cauchy–Schwarz gives $\sum_Q \sqrt{\mathbb{P}(Q)} \le 2^{jd/2}$, the choice $2^J \le n^{1/d} < 2^{J+1}$ yields $\mathbb{E}\mathcal{W}_1 \le 9\, d\, n^{-1/d}$ for $d \ge 3$. (ii) Concentration. Replacing one sample moves mass $1/n$ by at most $d$, so McDiarmid's inequality bounds the deviation by $d\sqrt{\log(1/\delta')/(2n)}$ with probability $1 - \delta'$. (iii) A union bound over two classes and $\binom{D}{d} \le (eD/d)^d$ subsets. For $d = 1, 2$ the first term becomes of order $n^{-1/2}$ and $n^{-1/2} \log n$ [10]. □

**Theorem 5.5** (Honest selection; proved given Lemma 5.4)**.** *Let* $\varepsilon$ *be as in Lemma 5.4 and* $\rho_k \ge \varepsilon_k$*. With probability at least* $1 - \delta$*, simultaneously for all* $|S| \le d$*:*

*(a)* $B_S \le V(S;\rho) \le V_{\mathrm{pop}}(S;2\rho)$*;*

*(b) the detector* $f_S$ *of the theory note at* $S$ *satisfies* $\mathbb{P}_0(f_S < 0) + \mathbb{P}_1(f_S \ge 0) \le R_{\mathrm{true}}(f_S) \le V(S;\rho)$.

*In particular the selected* $\widehat{S}$ *and its detector* $\widehat{f}$ *satisfy*

$$\mathbb{P}_0(\widehat{f} < 0) + \mathbb{P}_1(\widehat{f} \ge 0) \le R_{\mathrm{true}}(\widehat{f}) \le V(\widehat{S};\rho) \le \min_{|S|\le d} V_{\mathrm{pop}}(S;2\rho),$$

*and a selection that is optimal only up to* $\tau$ *satisfies the same chain with* $+\tau$ *at the right end.*

*Proof.* On the event of Lemma 5.4, Lemma 5.1 gives (a) because $\rho + \varepsilon \le 2\rho$, and Proposition 5.2 gives (b) because $\varepsilon_k \le \rho_k$. The chain follows from $V(\widehat{S};\rho) \le V(S;\rho)$ for all $|S| \le d$. □

The theorem says three things. The certificate stays valid after selection with no correction, because the event covers all subsets at once; and since validity holds for every $S$, exact optimization buys only the right end of the chain. The certificate never claims more separation than the Bayes risk allows, which removes the collapse of Table 2. And it is accurate up to doubling the radius. By Proposition 5.3, the order $d\,n^{-1/d}$ of the radius cannot be lowered for this detector class.

**Corollary 5.6** (Budget chosen by the certificate; proved given Lemma 5.4)**.** *For each* $d \le d_{\max}$ *let* $\rho(d)$ *be the radius of Lemma 5.4 at level* $\delta/d_{\max}$ *(or its analogue for* $d \le 2$*), and let* $(\widehat{d}, \widehat{S})$ *minimize* $V(S;\rho(d))$ *over* $d \le d_{\max}$ *and* $|S| \le d$*. With probability at least* $1-\delta$*,*

$$R_{\mathrm{true}}(\widehat{f}) \le V(\widehat{S};\rho(\widehat{d})) \le \min_{d\le d_{\max}} \min_{|S|\le d} V_{\mathrm{pop}}(S;2\rho(d)).$$

*Proof.* Apply Theorem 5.5 at each $d$ and take a union bound. □

The radius plays the role of the complexity penalty in structural risk minimization [1]: more coordinates separate the classes better but pay a larger radius, and the certificate balances the two without a separate penalty.

### 5.3 Rates and recovery

The right end of Theorem 5.5 is the Bayes risk plus the modulus $V_{\mathrm{pop}}(S;2\rho) - B_S$. Two hinge-loss examples fix its order. If $\mathbb{P}_0[S]$ and $\mathbb{P}_1[S]$ are supported on sets at $\ell_1$ distance at least $\Delta$, every unit of matched mass costs at least $\Delta$, so $V_{\mathrm{pop}}(S;r) \le 2(r_0+r_1)/\Delta$ and $B_S = 0$. If $\mathbb{P}_0 = U[0,1]$ and $\mathbb{P}_1 = U[1,2]$ on one coordinate, the cheapest way to match mass $m$ pairs $[1-m,1]$ with $[1,1+m]$ at cost $m^2$, so $V_{\mathrm{pop}}(S;r) = 2\sqrt{r_0+r_1}$ for $r_0 + r_1 \le 1$; smooth densities that cross behave alike, heuristically. With $\rho_k = \varepsilon_k$ and $s = |S^\star|$, the certified error is thus of order $s\,n^{-1/s}/\Delta$ in the first case and $(s\,n^{-1/s})^{1/2}$ in the second, up to the $\log(eD/s)$ term. The ambient dimension enters only through that logarithm, whereas $S = [D]$ would need a radius of order $D\,n^{-1/D}$.

**Proposition 5.7** (Recovery; proved given Lemma 5.4)**.** *Let* $|S^\star| = s$ *and suppose that* $V_{\mathrm{pop}}(S;\rho) \ge V_{\mathrm{pop}}(S^\star;\rho) + \gamma$ *for all* $S \ne S^\star$ *with* $|S| = s$*. If the radii of Lemma 5.4 at level* $s$ *satisfy* $\varepsilon_k \le \eta\rho_k$ *with* $4\eta/(1-\eta) < \gamma$*, then with probability at least* $1-\delta$ *the selection at budget* $s$ *returns* $S^\star$*.*

*Proof.* The map $t \mapsto V_{\mathrm{pop}}(S;t\rho)$ is concave, nondecreasing and takes values in $[0,2]$, so its slope on $[1-\eta, 1+\eta]$ is at most $2/(1-\eta)$. On the coverage event, Lemma 5.1 gives $V(S;\rho) \ge V_{\mathrm{pop}}(S;(1-\eta)\rho) \ge V_{\mathrm{pop}}(S;\rho) - 2\eta/(1-\eta)$ for every $S$, and $V(S^\star;\rho) \le V_{\mathrm{pop}}(S^\star;(1+\eta)\rho) \le V_{\mathrm{pop}}(S^\star;\rho) + 2\eta/(1-\eta)$. Hence $V(S;\rho) > V(S^\star;\rho)$ for every $S \ne S^\star$. □

Recovery thus needs $n_k$ of order $(Cs/\eta\rho)^s + s^3\log(eD/s)/(\eta\rho)^2$, with an absolute constant $C$ and $\rho = \min_k \rho_k$: exponential in the selected dimension, logarithmic in the ambient one, as in the power analyses of projected tests [33, 32].

### 5.4 Open problems

1. *Sharp radius.* The radius lies between $\frac{d}{4e}(4Mn)^{-1/d}$ (Proposition 5.3) and $c\,d\,n^{-1/d} + d^{3/2}\sqrt{\log(eD/d)/n}$ (Lemma 5.4). Is the $\log(eD/d)$ term necessary? A lower bound would study

the minimum of $V(S)$ over all $\binom{D}{d}$ subsets for identical classes, whose optimism the second row of Table 2 shows.

2. *Restricted detectors.* The order $n^{-1/d}$ is the price of the nonparametric class $\mathcal{F}_S$: the adversary of Proposition 5.3 defeats detectors that memorize isolated points. For linear or kernel detectors the variation analysis of [11] suggests radii of order $\sqrt{d \log(eD/d)/n}$, at the cost of the finite program of Lemma 2.1. The target is a version of Theorem 5.5 with such radii.

3. *Honest relaxation.* A weighted metric with weights $\zeta_m = d/D$ has, at scales below $d/D$, the covering numbers of a $D$-dimensional cube, so its honest radius is of $D$-dimensional order.

**Conjecture 5.8** (Honest relaxation)**.** *If each weighted metric in Proposition 4.1 is given the radius that makes it honest, the relaxation no longer improves on the best sparse selection by more than the width of the sandwich in Lemma 5.1. The gap of Section 4 would then be a statistical artifact.*

The first experiments are Table 2 with the radii of Lemma 5.4, recovery frequencies against $n$ for $D \in \{20, 100\}$, and a comparison with kernel selection [32].

# 6 Direction B: selection as matching interdiction

**Reference theory.** Interdiction games and bilevel integer programming [28], matching interdiction [35], interdiction with monotone followers [9], outer approximation [2], and the hardness of feature sets [6, 5].

**Route.** Program Eq. (2) is exact, and Proposition 4.2 says that a search tree cannot be avoided. The remaining gains must therefore come from the structure of the follower, which is a matching, and from a statistical reason to stop early.

**The game.** By Section 4, $\min_{|S|\le d} m_S(\rho)$ is a two-player game: the defender raises pair costs by choosing coordinates, and the adversary buys the largest cross-class matching it can afford. Removing arcs instead of pricing coordinates is matching interdiction [35]. The adversary's feasible set shrinks as coordinates are added, the monotone setting of [9]. Table 1 shows the limit of a generic solver: the tree grows almost as fast as enumeration, as Section 4 predicts.

**Proposition 6.1** (Matching bounds; proved)**.** *Let $\pi \ge 0$ satisfy the row and column constraints of Eq. (1), and let $\kappa_m = \sum_{i,j} \pi_{ij} C^m_{ij}$. For every $S$,*

$$m_S(\rho) \ge \sum_{i,j} \pi_{ij} \cdot \min\Big\{1, \frac{\rho}{\sum_{m\in S} \kappa_m}\Big\}.$$

*At a node of a search tree that fixes the coordinates $I$ in and leaves $J$ free, every completion satisfies the bound with $\sum_{m\in S} \kappa_m$ replaced by $\sum_{m\in I} \kappa_m$ plus the sum of the $d - |I|$ largest $\kappa_m$, $m \in J$.*

*Proof.* Scaling $\pi$ by the minimum keeps the row and column constraints and brings its cost under $S$ within $\rho$. The node bound follows because the bound decreases in $\sum_{m\in S} \kappa_m$. □

The bound charges a whole matching its exact cost under $S$, while the cuts of the theory note are affine in $z$. It costs $O(D \log D)$ per matching and node, and a pool of matchings collected from evaluated subsets gives the maximum of such bounds. Its strength is untested.

**Corollary 6.2** (Honest tolerance; proved given Lemma 5.4)**.** *If the search stops at absolute gap $\tau$ with the incumbent $\widehat{S}_\tau$, then on the event of Theorem 5.5, $R_{\text{true}}(\widehat{f}) \le V(\widehat{S}_\tau; \rho) \le \min_{|S|\le d} V_{\text{pop}}(S; 2\rho) + \tau$.*

*Proof.* Part (b) of Theorem 5.5 holds for every $S$, and $V(\widehat{S}_\tau; \rho) \le \min_S V(S; \rho) + \tau$. □

A natural $\tau$ is the width of the sandwich: with $\varepsilon_k \le \eta\rho_k$, the proof of Proposition 5.7 confines each $V(S; \rho)$ to an interval of length $4\eta/(1-\eta)$ around $V_{\text{pop}}(S; \rho)$, so smaller differences carry no statistical meaning. Tolerance alone does not save much, however. On the $D = 50$ instance of Table 1, relative tolerances of 10%, 5% and 2% stopped after 209, 211 and 220 s, against 235 s at the default tolerance, each run with the optimal subset; the solver finds the optimum early and spends its time raising the lower bound. These are single development runs.

**Proposition 6.3** (General losses; sketched)**.** *For any loss in the theory note, the problem*

$$\begin{aligned} \min \quad & \sum_{k=0}^{1}\Big(\frac{1}{n_k}\sum_i a_i^{(k)} + \rho_k\lambda_k\Big) \\ \textit{subject to} \quad & \ell((-1)^{k+1}g_y) \le a_i^{(k)} + \sum_m w_{km}|x_i^{(k)}[m] - y[m]|, \\ & 0 \le w_{km} \le \lambda_k, \qquad \rho_k w_{km} \le 2z_m, \qquad z \in \mathcal{Z}_d, \end{aligned}$$

*with rows over $k$, $i \in [n_k]$ and $y \in \mathcal{X}$, has the value* $\min_{|S|\le d} V(S)$.

*Proof sketch.* The diagonal rows give $a_i^{(k)} \ge 0$, and $g = 0$, $a = \mathbf{1}$, $\lambda = 0$ has value 2, so $\rho_k\lambda_k \le 2$ at an optimum. At $z = \mathbf{1}_S$ the choice $w_{km} = \lambda_k z_m$ is then feasible and optimal, which recovers the detector program of the theory note. □

For a piecewise-linear loss this is a mixed-integer linear program with $N^2$ rows and two prices; for the hinge loss, Proposition 3.2 reduces it to Eq. (2), with one price and $n_0 n_1$ rows.

**Open problems.** (i) Matching bounds and lazy generation of pair rows, starting from cross-class nearest neighbours, for large $n$. (ii) Certificate size: how many matchings are needed to prove optimality? Under the margin of Proposition 5.7, does a polynomial number suffice? This would be the computational counterpart of recovery. (iii) The first experiment replaces outer approximation by Eq. (2) on the 100-dimensional study.

# 7 Direction C: randomized selection

**Reference theory.** Linear programming duality for finite games, randomized defenses in adversarial classification [26, 18], feature-deletion games [13], and the problem of extending consistent marginals to a joint distribution [31].

**Route.** The $\ell_1$ cost splits across coordinates, so for every projection an attack is summarized by its coordinate-wise cost vector. A defender who mixes tangents of $V_S$ faces an attacker whose best reply spends the whole budget on one coordinate; linear programming duality turns the defender's problem into the attacker's budget allocation. For the value to be tight, the attacker must realize every projected least favorable pair at once, a marginal problem that is solvable when the candidate subsets are disjoint.

**The game.** Throughout this section $\rho = \rho_0 + \rho_1$ and $V_S(r) = 2m_S(r)$ is the hinge value of $S$ at total radius $r$. The defender draws a subset $S$ with $|S| \le d$ and a detector $f \in \mathcal{F}_S$ from a distribution $p$; the adversary chooses $(Q_0, Q_1) \in \mathcal{P}_0 \times \mathcal{P}_1$ knowing $p$ but not the draw. Let $\mathrm{OPT}_{\mathrm{rand}}$ be the defender's best worst-case expected risk. It is at most $\mathrm{OPT} = \min_{|S|\le d} V(S)$ and at least $\sup_Q \min_{|S|\le d}[2 - D_\ell(Q_0[S], Q_1[S])]$.

**Lemma 7.1** (Costs split; proved)**.** *For $(Q_0, Q_1) \in \mathcal{P}_0 \times \mathcal{P}_1$ reached by plans $\gamma_k$ within the budgets, let $r_m = \sum_k \int |x[m] - y[m]|\, d\gamma_k$. Then $\sum_m r_m \le \rho$, and for every $S$ the projected attack costs $\sum_k \mathcal{W}_1(\widehat{\mathbb{P}}_k[S], Q_k[S]) \le r(S) := \sum_{m\in S} r_m$.*

*Proof.* The $\ell_1$ cost is the sum of the coordinate costs, and projecting $\gamma_k$ gives a plan between the projections. □

By the proof of Proposition 3.3, each price $\lambda$ with witness $h$ gives a detector of $\mathcal{F}_S$ whose risk under an attack of projected cost $r$ is at most $A + \Lambda r$, with $A = 2(1-\beta_\lambda)$ and $\Lambda = 2\lambda$; and $V_S(r)$ is the minimum of these tangents over $\lambda$.

**Theorem 7.2** (Achievable randomized value; proved)**.** *Let $T$ be a finite family of tangents $(S_t, A_t, \Lambda_t)$. The best mixture over $T$ guarantees worst-case expected risk at most*

$$\min_p \max_{m\in[D]} \Big\{\sum_t p_t A_t + \rho \sum_{t:\, m\in S_t} p_t\Lambda_t\Big\} = \max_{r\ge 0,\ \mathbf{1}^\top r\le\rho} \min_{t\in T}\big\{A_t + \Lambda_t r(S_t)\big\} =: W_T.$$

*As $T$ grows to all tangents, $W_T$ decreases to $\overline{W} = \max_r \min_{|S|\le d} V_S(r(S))$, and* $\mathrm{OPT}_{\mathrm{rand}} \le \overline{W} \le \mathrm{OPT}$.

*Proof.* By Lemma 7.1, the expected risk of a mixture $p$ under any attack is at most $\sum_t p_t(A_t + \Lambda_t r(S_t))$ for the attack's cost vector $r$. This is linear in $r$ with nonnegative slopes, so its maximum over $\{r \geq 0, \mathbf{1}^\top r \leq \rho\}$ is attained at some $r = \rho e_m$, which gives the left side. The left side is the value of a finite bilinear game between the simplex and a polytope, so linear programming duality gives the equality. Adding tangents lowers $W_T$, and the minimum over all tangents of $S$ is $V_S(r(S))$; the convergence of the maxima follows from Dini's theorem on the compact set of budgets. Finally, $\overline{W} \leq \min_S \max_r V_S(r(S)) = \mathrm{OPT}$ because $V_S$ is nondecreasing and $r(S) \leq \rho$. □

The sum $\sum_{t:\, m \in S_t} p_t \Lambda_t$ is the price exposure of coordinate $m$. The attacker spends its whole budget on the most exposed coordinate, and the defender's mixture spreads exposure across coordinates; a fixed selection exposes its $d$ coordinates fully. This is "mixing helps the defender" in its strategic form, and it needs no minimax theorem beyond finite duality.

**Theorem 7.3** (Disjoint candidates; proved)**.** *If the candidate subsets are pairwise disjoint, in particular if $d = 1$, then $\mathrm{OPT}_{\mathrm{rand}} = \overline{W}$ with the minimum in $\overline{W}$ taken over the candidates, and $\overline{W}$ is given by water-filling: $\overline{W} = \max\{t : \sum_S V_S^{-1}(t) \leq \rho\}$, where $V_S^{-1}(t) = \inf\{r \geq 0 : V_S(r) \geq t\}$.*

*Proof.* The upper bound is Theorem 7.2. Since each $V_S$ is continuous and nondecreasing, an optimal budget $r^\star$ equalizes the candidates, which gives the water-filling formula. For the lower bound, take for each candidate $S$ a least favorable pair at total budget $r^\star(S)$, split between the classes in the ratio $\rho_0 : \rho_1$ (Proposition 3.2 allows any split). Lemma 2.1 of the theory note describes it by a plan from each source point. Move each source point independently in each candidate block according to these plans. The cost adds over blocks, so both classes stay within budget, and the projection on each $S$ is its least favorable pair. Every detector reading $S$ then has risk at least $V_S(r^\star(S)) \geq \overline{W}$, and so does every mixture. □

For example, take one point per class, $x^{(0)} = (0, 0)$ and $x^{(1)} = (1, 1)$, $\rho = 1/2$ and $d = 1$. Each coordinate has $V_{\{m\}}(r) = 2\min\{r, 1\}$, so the fixed optimum is 1, while water-filling gives $r^\star = (1/4, 1/4)$ and $\overline{W} = 1/2$. With $K$ disjoint candidates of the same separated type, the gain is a factor $K$. Table 3 shows the effect on the instances of Table 1: the randomized value is well below both the fixed optimum and the relaxation.

| $D$ | $N$ | $d$ | fixed $\min_S V(S)$ | randomized $W_T$ | relaxation |
|---|---|---|---|---|---|
| 10 | 30 | 1 | 1.78 | 1.24 | 1.53 |
| 10 | 30 | 2 | 1.40 | 0.59 | 0.89 |
| 15 | 30 | 2 | 1.12 | 0.41 | 0.75 |
| 12 | 30 | 3 | 0.87 | 0.32 | 0.49 |

Table 3: Instances of Table 1, $\rho_0 + \rho_1 = 0.6$. $W_T$ uses the tangents of each $V_S$ at 23 radii in $(0, \rho]$; it is achievable by Theorem 7.2, and for $d = 1$ it approximates $\mathrm{OPT}_{\mathrm{rand}}$ by Theorem 7.3. Development check.

The gain has a precise limit: it exists only against an adversary who commits to one attack with one shared budget.

**Proposition 7.4** (No gain against sampling error; proved)**.** *In the setting of Theorem 5.5, let a mixture $p$ use detectors with the certificates of Proposition 5.2. On the coverage event, the certificate of the mixture is $\sum_t p_t\big(A_t + \Lambda_t \sum_k \varepsilon_k\big)$, an average of fixed certificates, and hence no smaller than the best of them.*

*Proof.* Sampling error does not split a budget across coordinates: the coverage event allows every projection to be off by its full radius at once, so each term is bounded separately. □

Randomized selection is therefore a game-theoretic object. Its natural applications have a strategic adversary who must commit before the defender's coin, such as inspection, screening or sensor scheduling, and its guarantee holds on average over that coin.

**Open problems.** For overlapping candidates the attacker needs least favorable pairs that agree on the shared coordinates and whose coordinate costs respect one split of the budget. For decomposable families, pairs that agree can be glued into one attack [31]; for other families they need not be, and agreement itself may fail.

**Conjecture 7.5** (Tightness)**.** $\mathrm{OPT}_{\mathrm{rand}} < \overline{W}$ *on some instances when the candidates are all subsets of size* $d \geq 2$*; equality holds when the least favorable pairs at the optimal budget can be chosen to agree on intersections of a decomposable family.*

Computing $\overline{W}$ calls for column generation on the linear program of Theorem 7.2. Given the current budget vector $r$, the pricing problem is $\min_{|S| \leq d} V_S(r(S))$, which is Eq. (2) with $\rho\lambda$ replaced by $r^\top w$ and a bound on $\lambda$. General losses give two prices per tangent and otherwise the same construction.

# 8 Priorities and plan

1. *Honest radii* (Section 5) offer the strongest story at the lowest cost: selection is what makes a Wasserstein robust test valid in high dimension. Theorem 5.5 and Proposition 5.3 are proved up to the standard coverage lemma. Next come explicit constants for $d \leq 2$, Table 2 with honest radii, and recovery experiments.

2. *Matching interdiction* (Section 6) starts with Eq. (2) in the core package for the hinge loss, compared with outer approximation on the 100-dimensional study, then adds the matching bounds. The honest tolerance is free but saves little.

3. *Randomized selection* (Section 7) is exploratory. Its two theorems are proved; the marginal problem behind Conjecture 7.5 and a column-generation prototype come next, together with an application that has a strategic adversary.

Table 4 lists the status of each result. The routes of Section 2 that failed are not pursued. All numbers in this note come from small synthetic development checks (script dev_checks.py), not from formal experiments.

| Result | Status |
|---|---|
| Propositions 3.1, 3.2, 3.3: divergence, matching and witness forms | proved; 3.2, 3.3 checked numerically |
| Propositions 4.1, 4.2: compact program, hardness | proved |
| Lemma 5.1, Propositions 5.2, 5.3 | proved |
| Lemma 5.4: uniform coverage | standard; sketched |
| Theorem 5.5, Corollaries 5.6, 6.2, Proposition 5.7 | proved given Lemma 5.4 |
| Proposition 6.1: matching bounds | proved; untested |
| Proposition 6.3: general losses | sketched |
| Theorems 7.2, 7.3, Proposition 7.4 | proved |
| Conjectures 5.8, 7.5 | open |

Table 4: Status of the results in this note.

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